FGI-GSRx v2.1.2 — Release Notes

1. New Features in v2.1.2

1.1 RINEX file generation capability

FGI-GSRx v2.1.2 introduces the capability to generate navigation and observation RINEX 3.04 files from the processed data. Currently, file generation only supports GALILEO and GPS data. More specifically, navigation file generation supports GALILEO E1B & GPS C/A signals while observation file generation supports GALILEO E1B, GPS C/A, GPS L1C.

1.2 New Configuration Parameters

To enable RINEX file generation, users must configure <u>at least</u> the following parameters in their receiver configuration file (e.g., defaultReceiverConfiguration.txt, or any custom configuration file):

```
% RINEX parameters

rinex,enableRINEX, true,
fN,dir,"",                  % The path where you want to save the files
hD,fileType,[],             % 'O'=observation data, 'N'=navigation data, e.g.,  ['N' 'O']
…
```

Note that one must change the gpsRollovers parameter from the getSystemParameters.m to use data older than April 7th, 2019. The gpsRollovers parameter is set to 2 by default since there have been two rollovers since April 7th, 2019.

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2. Changes from v2.1.1 → v2.1.2

2.1 Added

\Folder\File	Remarks	Modification
\rinex \generateRinex.m	Main function of the RINEX file generation feature	**New file**
\rinex\writeRinex3Nav.m	Navigation file generation main function	**New file**
\rinex\writeRinex3Obs.m	Observation file generation main function	**New file**
\rinex\fileHandling\@rinex3*\cmps.m	Composition of RINEX 3.04 formatted header/data records	**New file**
\rinex\fileHandling\@rinex3*\isValid.m	Confirming data/header validity.	**New file**
\rinex\fileHandling\@rinex3*\rinex3*.m	Data type(s) to store data/header block information	**New file**
\rinex\fileHandling\@rinex3Obs*\parse.m	Parse functions to read rinex files	**New file**
\rinex\fileHandling\@rinex3NavData\setNav.m	Fill data object for one object	**New file**
\rinex\fileHandling\@rinex3ObsData\setObs.m	Sort and set observations for object	**New file**
\rinex\fileHandling\tna\carrierBand.m	Define carrier frequency bands	**New file**
\rinex\fileHandling\tna\obsType.m	Define observation types	**New file**
\rinex\fileHandling\tna\trackingMode.m	Define signal tracking modes	**New file**

\rinex\fileHandling\rinex3commonData.m	Common data type that nav/obs datatypes inherit. Stores common properties	**New file**
\rinex\fileHandling\rinex3commonHeader.m	Common data type that nav/obs headers inherit. Stores common properties	**New file**
\rinex\fileHandling\rinex3ObsId.m	Specify observation id based on tna	**New file**
\rinex\fileHandling\rinex3ObsMeas.m	Data object to store measurements of the observables	**New file**
\rinex\fileHandling\rinex3ObsTypeNum.m	Object to store observation types for a satellite system	**New file**
\rinex\fileHandling\rinex3PhaseShift.m	Object to store phase shift corrections	**New file**
\rinex\fileHandling\rinex3SatId.m	Object to store satellite id	**New file**
\rinex\fileHandling\satSysId.m	Define satellite systems	**New file**
\rinex\helperFunctions\generateFileName.m	Generate file names for the RINEX files	**New file**
\rinex\helperFunctions\getIonoCorr.m	Collect ionospheric correction information	**New file**
\rinex\helperFunctions\getMeasurements.m	Collectt obs measuremtnts for a given epoch	**New file**

\rinex\helperFunctions\getNavMeasurements.m	Collect nav measurements for given signal	**New file**
\rinex\helperFunctions\getSysObsNoTypes.m	Get an array of sorted rinex3ObsTypeNum objects corresponding to the obs types	**New file**
\rinex\helperFunctions\getSysPhaseShift.m	Generate system specific phase shift information for the defined obs types	**New file**
\rinex\helperFunctions\getTimeObs.m	Compute first, second and last observation captured by GSRx	**New file**
\rinex\helperFunctions\getTimesysCorr.m	Collect data for time system corrections	**New file**
\rinex\helperFunctions\getZeroTime.m	Calculate a valid zero time that accounts for GPS week rollovers	**New file**
\rinex\helperFunctions\inferSatSystem.m	Find correct satSysId for corresponding signal	**New file**
\rinex\helperFunctions\stna2fgiRep.m	Map rinex3ObsId to given signal names	**New file**

2.2 Modified

\Folder\File	Remarks	Modification
\frame\gpsl1DecodeEphemeris.m	Deletion of a manual gps week rollover correction	**Bug fix**
\main\gsrx.m	Addition of the rinex feature	**New feature**
\sat\getSatelliteInfo.m	Deletion of a manual gps week rollover correction	**Bug fix**
\track\pll\phaseFreqFilter.m	Introducing a new variable that counts the accumulated phase	**New variable**
\obs\getTransmitTime.m	Addition of accumulatedPhase to obs as accPhase. Tells the accumulated carrier phase for a given target sample count	**New variable**
\obs\generateObservations.m	Addition of accumulatedPhase to obsResults	**New variable**
\param\getSystemParameters.m	Addition of RINEX specific parameters	**New variables**